Jordan Mitchell

Selected Work

Mathematics / Assessment / Standards / AI Evaluation

50 examples of the work behind learning products: the standards maps, items, lessons, review systems, and editorial plans that make the final experience better for students and teachers.

50
selected pieces

K-16
math and learning

4
areas of work

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Selected work excerpts. Client-sensitive content has been redacted.

AI tutors and evaluation

10 pieces / What each one is and the part I played in it.

01 AI tutor evidence map

I mapped the evidence an AI tutor can use across grades 3–9, including patterns that signal a misconception. The map gives reviewers a concrete boundary for what the tutor can infer from student work.

05 AI-assisted item design

I set rules for question difficulty and answer explanations, then used examples to check whether generated items matched the intended standard.

14 AI assessment evaluation

I reviewed generated parent and child items for alignment, clarity, readability, and answer quality, and recorded the issues for revision.

21 Student conversation review

I reviewed seven selected tutor cases to test which evidence about averages and range could be identified reliably. The resulting notes distinguish a student error from a tutor inference.

22 Grade 1 mission specification

I defined how each scored mission connects to one assessable component while keeping prerequisites and grade-level mastery separate.

23 Cross-course learning objective map

I directed a map of objectives shared across courses, making repetition and gaps visible before content was revised.

24 Learning objective verifier

I helped set up a check that compares chapter-map objectives against both slides and worksheets, then flags where each one appears.

32 DOK 4 tutor-task prompt

I set rules for an extended-thinking task a student can complete with an AI tutor. The prompt asks for decisions, testing, revision, and reflection while staying inside the grade-level standard.

33 Math question gap-filling prompt

I wrote guardrails for generating standards-aligned math questions when a bank has a gap. The questions have to use realistic quantities, clear units, and enough information to stand on their own.

47 Ratios and percents tutor evidence map

I mapped productive and problematic reasoning about ratios and percents using 31 anchor questions and 95 eligible tutor conversations. The evidence sets boundaries for what the tutor can infer.

Standards and assessment

20 pieces / What each one is and the part I played in it. / Part 1 of 2

02 Math standards crosswalk

I worked through a crosswalk covering 2,194 questions and made the quality of each standards match visible for review.

03 Question bank coverage audit

I checked the approved question bank against 148 standards and documented 43 gaps for targeted writing.

06 Florida alignment review

I reviewed a standards-mapping output against lesson activities and recorded where the match was exact, partial, or needed correction.

07 Texas TEKS correlations

I supervised a correlation across two higher-ed math titles and documented where the Texas standards were covered or missing.

09 Kindergarten selected-response practice

I helped move six missions of standards-aligned questions through a fast draft and revision cycle, with answer rationales for each item.

10 TX STAAR assessment review

I wrote a review checklist for TX STAAR math items covering wording, math, metadata, and accessibility. The 900-item program reached 98% no-rewrite acceptance.

11 Geometry benchmark items

I wrote standards-tagged geometry items with DOK levels and feedback tied to likely student mistakes.

12 Adaptive statistics questions

I wrote adaptive statistics questions with worked answers and explanations that address common errors.

13 Calculus test-bank system

I built a parent-item template for consistent variants, then drafted calculus questions with solution and distractor notes.

20 Applied calculus questions

I wrote application questions that connect calculus to real situations and show each step of the solution.

31 Five-grade standards alignment prompt

I wrote a five-grade check for deciding whether a math question truly fits a target standard. It tests number limits and representations before looking at nearby grades.

34 Multiple-integrals test-bank blueprint

I planned six multiple-integrals lessons by exercise family, DOK level, objective, and graphic need. The blueprint organized 48 exercise sets and 528 questions for writing.

Standards and assessment

20 pieces / What each one is and the part I played in it. / Part 2 of 2

35 Calculus item math and dependency audit

I audited a 25-question calculus bank for math, answer logic, LaTeX, and variant dependencies. The review calls out six blocking errors and gives writers exact fixes.

36 Precalculus quiz review log

I tracked changes to 17 precalculus quiz questions through QC and SME review. The log shows which feedback on number lines, notation, and solution steps was addressed.

37 Piecewise-function item repair

I revised a 27-item piecewise-function set to make graph choices and endpoint inclusion precise. The excerpt shows how the answer options differ at the boundaries.

38 Algebra II and geometry assessment blueprint

I helped plan 30-item geometry and algebra forms, balancing item types and mathematical domains so the assessment covers more than routine multiple choice.

40 Taylor-series item review

I reviewed higher-ed calculus items on binomial and Taylor series, including learning objectives and answer choices. The excerpt shows the section structure and the questions under review.

41 Florida B.E.S.T. to CCSS crosswalk

I compared grades 3–5 B.E.S.T. benchmarks with CCSS standards and recorded the strength and limits of each match. The notes make the alignment decision visible.

42 Financial Algebra quality-rubric review

I checked a Financial Algebra program against a Texas instructional-materials rubric. The evidence log ties specific course materials to each review criterion.

43 Grade 4 standards remapping audit

I reviewed 386 flagged grade 4 rows where the assigned standard changed. The workbook preserves the tested skill, old code, new code, and rationale.

Curriculum and instruction

11 pieces / What each one is and the part I played in it.

04 Grades 1–6 skill tracker

I broke six grades of math standards into assessable skills so the team could plan coverage without counting the same skill twice.

08 Teacher resource guides

I led the development and review of teacher guides for statistics and quantitative reasoning. A prototype led to full guides for two titles.

15 Interactive math video

I reviewed a video lesson that pairs explanation with questions about data spread so students work through the comparison themselves.

16 Worked examples in math

I developed and reviewed short worked-example visuals that show two ways to find the area of a rectangle, from counting squares to multiplying dimensions.

17 Higher-ed statistics worksheets

I reviewed higher-ed math practice across a large worksheet program, including questions on confidence intervals and margins of error.

18 Secondary math course practice

I reviewed the math and teaching in a secondary teacher-preparation course, including practice on modeling real constraints with inequalities.

19 Comparing decimals lesson

I built a model lesson that uses thermometers and number lines to make decimal comparison and rounding concrete.

39 Residual-plot student activity

I helped shape a statistics activity that moves from a scatterplot to a regression line and residual plot. Students use the evidence to decide whether a new point is influential.

45 Grade 7 circle-graph lesson sample

I provided editorial and math review on a grade 7 lesson that moves from reading a circle graph to using proportions and sector angles. The excerpt shows the student-facing progression.

46 Introducing multiplication unit design

I helped guide a multiplication unit from equal groups and arrays through skip counting and word problems. The sequence shows how early concepts build before independent practice.

48 Expected-value student activity

I reviewed an expected-value activity built around an extended-warranty decision. Students use probabilities and payouts to explain what the expected value means for a consumer.

Editorial leadership and operations

9 pieces / What each one is and the part I played in it.

25 Teacher guide delivery case study

I documented how the team kept a teacher-guide project moving through a two-month pause. The successful prototype led to completed guides for two math titles.

26 Time-on-task estimating model

I broke teacher-guide work into repeatable parts so the schedule and effort estimate could scale with the scope.

27 Assessment authoring workflow

I planned writer assignments, item volume, and review timing for a large calculus assessment program.

28 Math team capabilities deck

I presented the team's range of math work, from assessment writing and standards alignment to video and AI-content review.

29 Canadian localization QA

I directed a shared terminology list and QA checkpoints so localized lessons stayed consistent across courses.

30 Water distribution math review

I checked a technical training proof against its answer key and recorded six specific corrections to units, wording, and calculations.

44 Math-content census across technical titles

I inventoried pages with math across technical training books so the team could size its review effort. This is the page-level count behind the staffing estimate.

49 Math alt-text rules

I helped guide the rules for describing equations, tables, and graphs in math content. The examples show how spoken math can stay precise without overloading the reader.

50 Lesson production and review schedule

I scheduled 96 lessons across seven freelancers, from parent-item drafting through review, variants, and final approval. The plan makes each handoff and its volume visible.

AI tutor evidence map

My role Math content and evaluation design

I mapped the evidence an AI tutor can use across grades 3–9, including patterns that signal a misconception. The map gives reviewers a concrete boundary for what the tutor can infer from student work.

Purpose and boundary

This document defines the productive and problematic evidence targets for Map 35 across Grades 3-9. It is an evidence-extraction reference, not the deterministic belief-state model itself.

Core principle: The LLM identifies student evidence and its support context. The deterministic model decides how that evidence updates facet beliefs.

Cross-grade progression

Grade band	How the evidence becomes more demanding
Grades 3-4	Relational meaning of equality; open sentences; unknowns; representing simple situations; keeping compared quantities on opposite sides.
Grades 5-6	Variables as quantities; substitution; one-step equations; multiplicative relationships; transition from arithmetic reasoning to generalized relationships.
Grades 7-8	Multi-step equations and inequalities; full solution sets; boundary inclusion; inequality direction; algebraic equivalence and symbolic modeling.
Grade 9	Literal equations, linear relations, and simple systems, while retaining the same foundational equality, variable, inverse-operation, and balance facets.

Progression rule: The facets do not reset at each grade. Later items combine earlier facets in longer, less familiar, or more abstract situations.

Math standards crosswalk

My role Standards mapping and review leadership

I worked through a crosswalk covering 2,194 questions and made the quality of each standards match visible for review.

WORKBOOK EXCERPT / Summary

Math standards crosswalk — reviewed sample	
Rows (all math questions)	2194
Grades	G3: 235 G4: 427 G5: 480 G6: 207 G7: 393 G8: 258 G9: 194
Evidence quality	EXACT: 399 CLOSE: 932 TOPIC: 274 NONE: 589
Row colors	green = corrected in row-by-row review; yellow = needs review (content); orange = needs review (figure unavailable); white = verified
Review status	VERIFIED: 1305 CORRECTED: 577 NEEDS REVIEW - content: 107 NEEDS REVIEW - visual unavailable: 205
Rows individually re-reviewed (batches 1-7, from bottom)	350
Second-review findings integrated (top-section rows)	608 rows: 314 standards adopted, 283 confirmed, 7 held with responses; all 20 comments answered in the Response column

Question bank coverage audit

My role Coverage analysis

I checked the approved question bank against 148 standards and documented 43 gaps for targeted writing.

WORKBOOK EXCERPT / Start Here

Measure	Count
Standards reviewed	148
Standards with confirmed gaps	43
Coverage-gap rows (one per standard)	43
Separate mapping fixes	12
Approved Anchor/Bank questions reviewed	1866

Grades 1–6 skill tracker

My role Curriculum and skill decomposition

I broke six grades of math standards into assessable skills so the team could plan coverage without counting the same skill twice.

WORKBOOK EXCERPT / Summary

[redacted] Curriculum Tracker			
Common Core Mathematics standards and assessable skills for Grades 1–6.			
Grade	Standards	Assessable Skills	Avg. Skills per Standard
Grade 1	21	85	4.04761904761905
Grade 2	26	112	4.30769230769231
Grade 3	25	119	4.76
Grade 4	28	123	4.39285714285714
Grade 5	26	98	3.76923076923077
Grade 6	29	121	4.17241379310345
TOTAL (Grades 1-6)	155	658	4.24516129032258

AI-assisted item design

My role Prompt and assessment review

I set rules for question difficulty and answer explanations, then used examples to check whether generated items matched the intended standard.

ORIGINAL DOCUMENT EXCERPT

Model: ChatGPT 5, since it has searching capabilities I have had best results with it understanding standards better. I also had to reprompt it to use grade appropriate words, since it was spitting out eg 'botanist' for a 2rd grade question. Question Prompt: Same as dev_prompts/erb_qgen_step1_math.txt but asked it to tag it to the closest CCSS, FL, and Arizona standards as well. Answer explanation prompt: Write an “Answer explanation” only. Do not restate the question. Use clear, plain English at about a reading level matching the standard. Use third-person and active voice. Keep it basic and concise.

Use the wording of the standard in the explanation. Prefer short sentences that follow this flow: identify the rule or pattern in words; state the goal; write a number sentence or equation if applicable; substitute known values; solve for the unknown; confirm the result fits the rule or equality; end with a final sentence that gives the result in context.

Keep unnecessary wording and jargon out. Do not use a two-column format. Present the reasoning in a simple sequence of sentences and math. Easy = DOK 1: The student recalls or recognizes facts, terms, or simple procedures and carries them out exactly as learned.

Medium = DOK 2: The student uses skills and concepts to make choices, organize or summarize information, and explain relationships in one or two steps.

Hard = DOK 3: The student thinks strategically by planning, reasoning, and justifying answers with evidence across multiple steps or solution paths.

Question:

Florida alignment review

My role Product and SME review

I reviewed a standards-mapping output against lesson activities and recorded where the match was exact, partial, or needed correction.

WORKBOOK EXCERPT / Sheet1

Lesson Title	Activity Type	Florida Standards	US review of standards	Mapping (Partial/No mapping/Completely Mapping)
Modeling Addition and Subtraction				
	Activate	MA.K.NSO.2.1	This is only partial because it's an intro to the full standard so I think that's ok. Currently it	partial
	Explore	MA.1.NSO.2.1	MA.1.NSO.2.1: Exact MA.1.NSO.2.2: Partial	partial
	Explore	MA.1.AR.1.1	MA.1.NSO.2.1: Exact MA.1.NSO.2.2: Partial	partial
	Reflect	MA.1.AR.2.1	MA.1.NSO.2.1: Exact MA.1.NSO.2.2: Partial	partial

Texas TEKS correlations

My role Supervising editor and case-study writer

I supervised a correlation across two higher-ed math titles and documented where the Texas standards were covered or missing.

WORKBOOK EXCERPT / TEKS Breakout Correlations

TEKS Knowledge and Skills Statement	Breakout	Item Type	Page(s)	Description of the specific location
(1) Mathematical process standards. The student uses	(i) apply mathematics to problems arising in everyday	Activity/Assessment	219	Linking Concepts #1-2
		Activity/Assessment	243	Viewpoint
		Activity/Assessment	440	Problem 3
		Activity/Assessment	79	Writing Projects #4
		Narrative	174	First paragraph until example 1
		Narrative	180	Fourth paragraph
		Narrative	505-506	Guided Exercise 5
		Narrative	44-45	Guided Exercise 1
(1) Mathematical process standards. The student uses mathematical processes to	(ii) apply mathematics to problems arising in society	Activity/Assessment	263	Using Technology

Teacher resource guides

My role Supervising editor

I led the development and review of teacher guides for statistics and quantitative reasoning. A prototype led to full guides for two titles.

ORIGINAL DOCUMENT EXCERPT

Chapter 1

Chapter Overview

SOURCE STRUCTURE / CONTENT WITHHELD

01	02	03	04
Chapter overview	**Scope and sequence**	**Key terms**	**Guided practice**

Teacher guide outline. Selected work excerpt; client-sensitive content redacted.

Kindergarten selected-response practice

My role Supervising editor

I helped move six missions of standards-aligned questions through a fast draft and revision cycle, with answer rationales for each item.

ORIGINAL DOCUMENT EXCERPT

Which shape could belong to both groups?

Question 1 (K.2A)

Correct Answer: A

To find the missing number, students must either count up one from 2, or down one from 4. Selecting 3 shows an understanding of counting forward or backward.

Incorrect (B):

Student counted up from 4, suggesting a misunderstanding of the order of numbers.

Incorrect (C):

Student counted the total number of stars, indicating a misunderstanding of the problem structure.

Question 2 (K.2B)

Correct Answer: C

The image shows 5 stickers, meaning the student can represent whole numbers with objects.

Incorrect (A):

The picture shows 2 stickers, indicating a confusion between the number 2 and the number 5.

TX STAAR assessment review

My role Editor and checklist author

I wrote a review checklist for TX STAAR math items covering wording, math, metadata, and accessibility. The 900-item program reached 98% no-rewrite acceptance.

ORIGINAL DOCUMENT EXCERPT

Editor's Checklist.

Open the review tracker and locate the item by ID.

Use the question ID to open the item in the review system.

Click on the eye to preview the item.

Check the item text is worded grammatically correct and clearly conveys a question that assesses the objective. You may have to adjust the wording to make sure it's coming across clearly. You can check the objective by clicking the arrow on the top right of an item.

Check the answer explanation on the bottom of an item, after pressing open. Ensure that the answer explanation is in accordance with the AE document: Answer Explanation Vendor Facing Authoring Guidelines.pdf

Ensure that writers use LaTeX for any math symbols that cannot be typed on a keyboard. Additionally, decimals should be entered using LaTeX. Additionally, if LaTeX is used at all in the item, (i.e. there is a decimal number or equation) then all numbers, variables, mathematical symbols used within that question are to be entered using LaTex, even if they can be typed using a keyboard.

Make sure any art go-bys are accurate and contain alt text that succinctly and accurately describes the art without giving away any additional information.

Check the metadata on the item:

Geometry benchmark items

My role Item author and editor

I wrote standards-tagged geometry items with DOK levels and feedback tied to likely student mistakes.

ORIGINAL DOCUMENT EXCERPT

Item metadata | Reference ID withheld; Assessment: BOY; Grade: 10;
CCSS Standard: G.CO.3

Item Type | Multiple Choice

Compose question | Rectangle ABCD is centered at the origin. ; ; ; ; Which transformation maps the figure onto itself?

MC Options | 180° rotation about the origin; 90° counterclockwise rotation about the origin; Reflection across the line y = x; Reflection across the

MC Distractor Level Feedback | Correct; Requires rectangle ABCD to be a square.; Requires rectangle ABCD to be a rhombus.; Requires rectangle ABCD to be a rhombus.

Correct answer | 180° rotation about the origin

Distractor Rationale |

Adaptive statistics questions

My role Math item writing and review

I wrote adaptive statistics questions with worked answers and explanations that address common errors.

ORIGINAL DOCUMENT EXCERPT

SOURCE STRUCTURE / CONTENT WITHHELD

01

Learning objective

02

Adaptive question

03

Worked answer

04

Error explanation

Question framework. Selected work excerpt; client-sensitive content redacted.

Calculus test-bank system

My role Template and item design

I built a parent-item template for consistent variants, then drafted calculus questions with solution and distractor notes.

ORIGINAL DOCUMENT EXCERPT

SME Parent Item Draft Template

Project: Early Transcendental Test Bank

Brief SME instruction: Complete one copy of this template for each parent item. Use the dropdowns where provided. Delete optional sections that do not apply before submitting the item for review. Keep math notation in the format requested by the project lead.

1. Item Metadata

2. Stimulus / Question Stem

Include all information the student needs, including any directions, context, data, equations, graph references, or table references.

3. Answer Content

Delete what isn't needed.

Multiple Choice / Multiple Select Answer Options

For non-MC items, delete this table. Keep each rationale aligned to the same error pattern across variants.

Fill-in-the-Blank / Numeric Entry / Free Response Answer

For MC or MS items, delete this table if it is not needed.

4. Worked Solution

AI assessment evaluation

My role Review criteria and calibration

I reviewed generated parent and child items for alignment, clarity, readability, and answer quality, and recorded the issues for revision.

WORKBOOK EXCERPT / Data

Item Code	Item Alignment Issue	Clarity	Necessary information	Readability
Item #: ES1_P1	No issue	Not concisely stated	Lacks some information	Not easily readable
Item #: ES1_P1_C1	No issue	Concisely stated	Contains all information	Easily Readable
Item #: ES1_P1_C 2	No issue	Not concisely stated	Lacks some information	Easily Readable
Item # ES1_P2	No issue	Concisely stated	Contains all information	Easily Readable
Item # ES1_P2_C1	No issue	Concisely stated	Contains all information	Easily Readable
Item # ES1_P2_C2	No issue	Concisely stated	Contains all information	Easily Readable
Item # ES1_P2_C3	No issue	Concisely stated	Contains all information	Easily Readable
Item # ES1_P3	No issue	Concisely stated	Contains all information	Easily Readable
Item # ES1_P3_C1	No issue	Concisely stated	Contains all information	Easily Readable

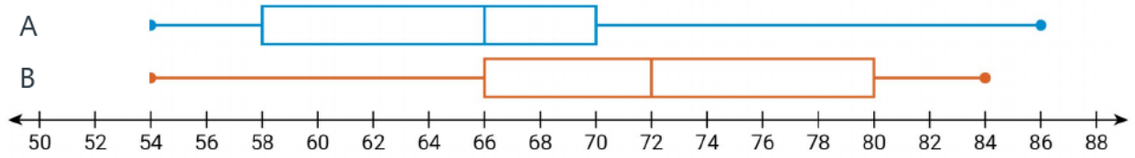
Interactive math video

My role Editorial and SME direction

I reviewed a video lesson that pairs explanation with questions about data spread so students work through the comparison themselves.

A manager compares lunchtime customer counts at two locations over 11 days. Compare the IQR for the data sets.

A	56	62	66	57	54	66	68	59	72	86	72
B	58	54	67	72	78	82	77	65	71	84	84



The IQR for location A is _____.
The IQR for location B is _____.
Which location has more variable customer counts?

Interactive lesson excerpt. Selected work excerpt; client-sensitive content redacted.

Worked examples in math

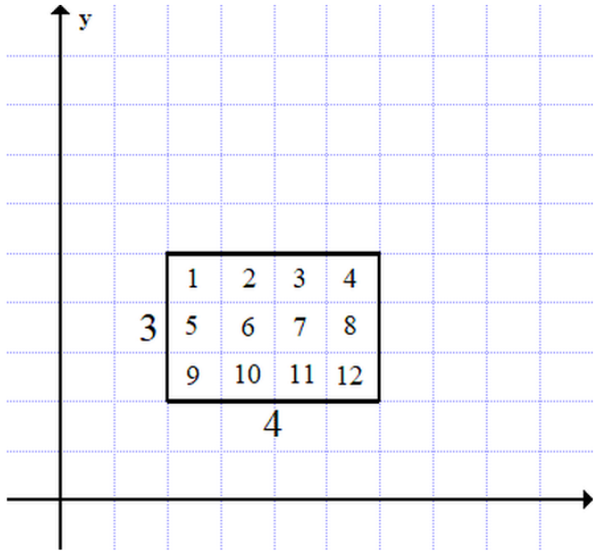
My role Slide development and editorial direction

I developed and reviewed short worked-example visuals that show two ways to find the area of a rectangle, from counting squares to multiplying dimensions.

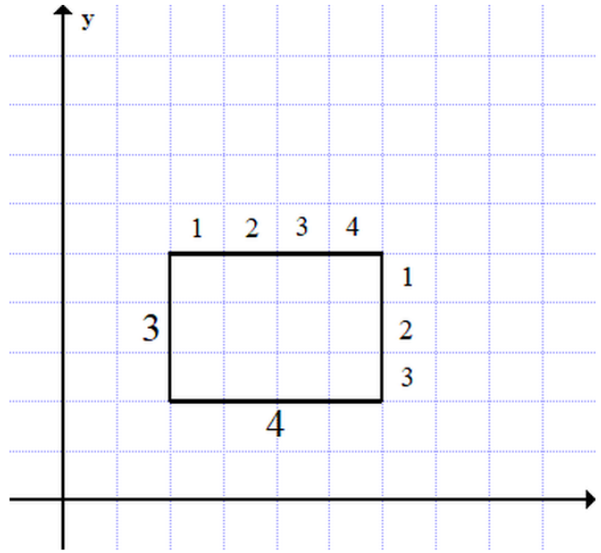
ORIGINAL SLIDE ART AND TEXT

A rectangle has a height of 3 units and a width of 4 units. What is its area?

Counting Squares to Determine Area



Multiplying to Determine Area



Higher-ed statistics worksheets

My role Editorial and SME review

I reviewed higher-ed math practice across a large worksheet program, including questions on confidence intervals and margins of error.

ORIGINAL DOCUMENT EXCERPT

Calculate a margin of error given a confidence interval

Suppose we know that a confidence interval for the mean height, in inches, of a population is (48.5, 56.8). Find the margin of error.

Suppose we know that a confidence interval for a population proportion is (0.37, 0.64). Find the margin of error.

Suppose we know that a confidence interval for the mean price of all textbooks (in dollars) sold at the campus bookstore is (110,195). Find the margin of error.

Suppose we know that a confidence interval for the average weight of a school supply box for the kindergarteners (in ounces) is (99,110). Find the margin of error.

A coach sampled student athletes about whether they ate a snack before practice.
The resulting confidence interval is (0.69, 0.85). Find the margin of error.

The weights, in pounds, of the dogs at a dog park for small dogs are normally distributed. If a random sample of dogs is taken and the confidence interval is (8.5, 24.6), what is the margin of error?

Statistics worksheet excerpt. Selected work excerpt; client-sensitive content redacted.

Secondary math course practice

My role Supervising editor and SME review

I reviewed the math and teaching in a secondary teacher-preparation course, including practice on modeling real constraints with inequalities.

ORIGINAL DOCUMENT EXCERPT

Guided Practice

Multiple Select

A park is planning a tree-planting event and has two types of trees to plant: Oak trees (T) and Pine trees (P). The total number of trees planted cannot exceed 400. Additionally, each Oak tree requires 5 gallons of water per week, and each Pine tree requires 3 gallons of water per week. The park has a weekly water supply of 1,500 gallons for the trees.

Which of the following inequalities accurately represent the constraints in the tree-planting scenario? Select all that apply.

Correct answer: $T + P \leq 400$

Rationale: This inequality correctly represents the limit on the total number of trees that can be planted.

Correct answer: $5T + 3P \leq 1{,}500$

Rationale: This inequality correctly represents the limit on the amount of water available for the trees.

Distractor: $T + P \geq 400$

Rationale: This inequality incorrectly suggests that at least 400 trees must be planted, which is not stated in the problem.

Distractor: $5T + 3P \geq 1{,}500$

Rationale: This inequality incorrectly suggests that at least 1,500 gallons of water must be used, which is not stated in the problem.

Comparing decimals lesson

My role Lesson author and editor

I built a model lesson that uses thermometers and number lines to make decimal comparison and rounding concrete.

ORIGINAL DOCUMENT EXCERPT

Lesson 4 Compare, Order, and Round Decimals

ESSENTIAL QUESTION How can a number line help compare and round decimals?

[Insert photo of a set of three digital lab thermometers shows readings of 4.198 °C, 4.891 °C, and 4.918 °C.] Look at the picture of three thermometers.

Write the three temperatures from least to greatest. _____, _____, _____

Explain in one sentence how you decided. _____

_____

Draw a horizontal number line that starts at 4 °C and ends at 5 °C.

Mark each tenth between 4 and 5 with a longer tick. Approximate where each point should be based on their tenths digit and plot them.

Finally, list the numbers in ascending order. The point farthest to the left has the least value and the one on the right has the greatest value. Guided Practice 1

A student streamed two science videos, one 4.5 minutes long and one 4.2 minutes long.
Which video is longer?

To answer this question, you can create a number line between 4 and 5.

Applied calculus questions

My role Item author and editor

I wrote application questions that connect calculus to real situations and show each step of the solution.

ORIGINAL DOCUMENT EXCERPT

Question:

A startup models its revenue as R(n) = n^2 million dollars when selling n thousand units. Use differentials to approximate the change in revenue when sales increase from n = 1.00 to n = 1.05 (in thousands).

Student Response box:

ΔR ≈ _____ (million dollars)

Answer:

0.10

Answer explanation:

Treat dR = R′(n) dn. Here R′(n) = 2n. At n = 1, R′(1) = 2. The change in n is dn = 0.05, so ΔR ≈ 2 × 0.05 = 0.10 million dollars.

Module URL and identifier withheld

Learning Objective Description: Investigate linear approximations and applications

Concept: Investigate linear approximations and applications

Atom Header: Text Instructional

Student conversation review

My role AI tutor math evaluation

I reviewed seven selected tutor cases to test which evidence about averages and range could be identified reliably. The resulting notes distinguish a student error from a tutor inference.

ORIGINAL DOCUMENT EXCERPT

Case | Validation impact

SOURCE STRUCTURE / CONTENT WITHHELD

01	02	03	04
Tutor case	**Observed reasoning**	**Evidence label**	**Validation decision**

Grade 1 mission specification

My role Curriculum and product design

I defined how each scored mission connects to one assessable component while keeping prerequisites and grade-level mastery separate.

ORIGINAL SPECIFICATION EXCERPT

Normalization rules

One Home Standard:** Each atomic assessable component has one canonical home standard.

One Primary Component Per Scored Mission:** Each scored mastery mission has exactly one primary assessable component. Supporting/prerequisite skills may be used but do not receive new mastery evidence from that mission.

Standard Mastery Floor:** At least 2 qualifying DOK 1, 2 qualifying DOK 2, and 2 qualifying DOK 3 missions per standard.

Coverage Rule:** Every required assessable component must receive qualifying evidence. Therefore minimum scored missions per standard = max(required component count, 6), before adding replacement/parallel forms.

Representation Rule:** Objects, drawings, equations, and tools are classified as representations/supports unless the standard explicitly makes the representation or explanation itself part of the requirement.

No Duplicate Skill Ownership:** A skill used by another standard is referenced as a prerequisite/support rather than copied as a new component.

Entry Assumption:** The main Grade 1 path assumes only recognition of numerals 0-9. Kindergarten prerequisites not assumed at entry must be taught/checked in a bootstrap support layer rather than counted as Grade 1 mastery components.

Dok Note:** DOK is assigned at the mission level, not permanently to a component. The same component can be assessed at different DOK levels through different gameplay demands.

Cross-course learning objective map

My role Editorial direction

I directed a map of objectives shared across courses, making repetition and gaps visible before content was revised.

WORKBOOK EXCERPT / Summary

Domain	Shared LOs in domain	Courses in domain	Max courses for 1 LO
Applied Mathematics	1077	10	24
Biology	76	2	2
Calculus	1746	9	24
Chemistry	792	11	8
College Algebra & Precalculus	1141	16	24
College Mathematics	2698	14	24
Contemporary Mathematics	170	2	24
Economics	758	8	21
Foundations of Math / Algebra	1206	7	24
Physics	442	2	22

Learning objective verifier

My role Tool direction and verification

I helped set up a check that compares chapter-map objectives against both slides and worksheets, then flags where each one appears.

ORIGINAL VERIFICATION REPORT EXCERPT

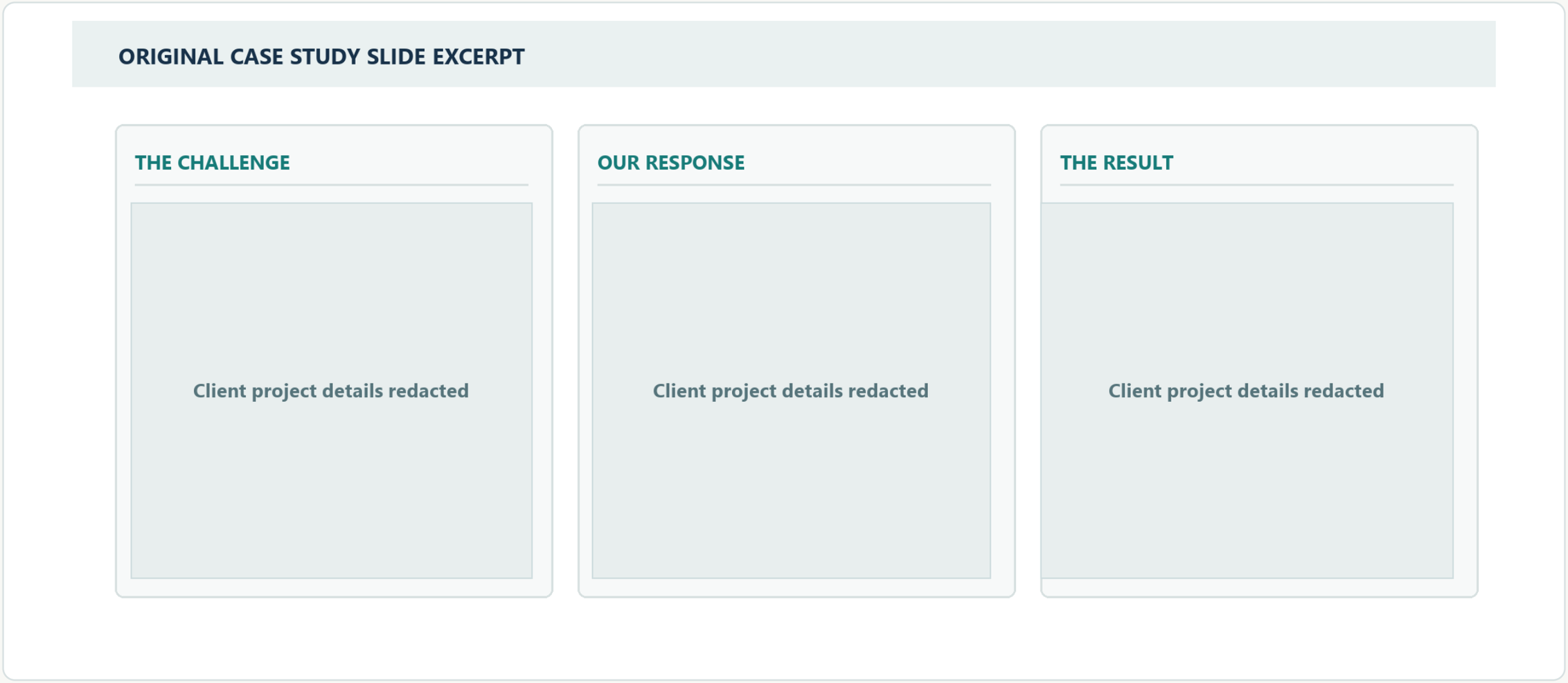
LO	Overall	PptWhere	WsWhere
Distinguish between natural numbers, whole numbers, and integers	OK - exact in both	Slides 2, 7-14	Page 1
Distinguish between rational and irrational numbers	OK - exact in both	Slides 2, 15-18	Pages 1-2
Perform calculations using order of operations	OK - exact in both	Slides 2, 19-23	Pages 1-2
Use the inverse and identity properties of real numbers	OK - exact in both	Slides 2, 24-30	Pages 1-2
Use the commutative, associative, and distributive properties of real numbers	OK - exact in both	Slides 2, 31-36	Pages 1, 3
Evaluate algebraic expressions with a single variable	OK - exact in both	Slides 3, 37-41	Page 3
Evaluate algebraic expressions with two variables	OK - exact in both	Slides 3, 42-45	Page 3

Verification report excerpt. Selected work excerpt; client-sensitive content redacted.

Teacher guide delivery case study

My role Case study writing and project leadership

I documented how the team kept a teacher-guide project moving through a two-month pause. The successful prototype led to completed guides for two math titles.



Case study slide. Selected work excerpt; client-sensitive content redacted.

Time-on-task estimating model

My role Estimating and project planning

I broke teacher-guide work into repeatable parts so the schedule and effort estimate could scale with the scope.

WORKBOOK EXCERPT / Lesson-Level

Element	Total Occurrences	Average per Lesson	Estimated Time per Element (minutes)	Total Time for 40 Lessons (minutes)
Lesson Overview	40	1	30	1200
Differentiated Instruction*	44	1.1	30	1320
Teaching Tips*	22	0.55	10	220
Discussion Questions*	44	1.1	10	440
Focus Questions*	44	1.1	10	440
Class Activities*	22	0.55	20	440
Writing Prompts*	22	0.55	20	440
Additional Examples	40	1	20	800

Assessment authoring workflow

My role Production planning and editorial direction

I planned writer assignments, item volume, and review timing for a large calculus assessment program.

WORKBOOK EXCERPT / Freelancer Assignments

Freelancer	Lesson Count	Parent Questions	Variants	Total Items
Freelancer A	20	500	3000	3500
Freelancer B	25	625	3750	4375
Freelancer C	25	625	3750	4375
Freelancer D	26	650	3900	4550

Math team capabilities deck

My role Team and service direction

I presented the team's range of math work, from assessment writing and standards alignment to video and AI-content review.

ORIGINAL SLIDE CONTENT

Math Content Authoring and Editing (continued)

Authoring math items aligned to state assessments, GED, Praxis, and AP-style exams

Scripting project-based math tasks and multi-step performance items

Developing scripts, creating video lessons in a wide range of formats including instructor led, animated, AI-generated, and delivering final, accessible files

Building educator resources, including pacing guides and teacher resource guides

Writing student-facing study tools such as checklists and reference sheets

Formatting and updating glossaries, indexes, and tables of contents

Creating and/or reviewing AI-generated math content for logic, correctness, and test readiness

Creating, reviewing, or editing student- and teacher-facing materials for content, accuracy, and alignment

Canadian localization QA

My role Editorial workflow direction

I directed a shared terminology list and QA checkpoints so localized lessons stayed consistent across courses.

WORKBOOK EXCERPT / Spelling Changes

US Spelling	Canadian Spelling	Notes
analog	analogue	
behavior, behavioral	behaviour, behavioural	
caliber	calibre	
catalog	catalogue	
center, centered	centre, centred	
check (financial)	cheque	
color, colorful	colour, colourful	
counselor	counsellor	
credentialing	credentialling	
defense	defence	

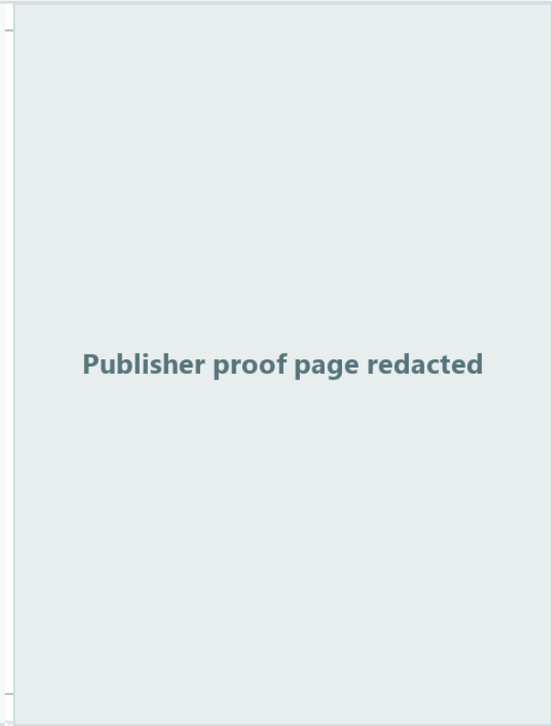
Terminology workbook excerpt. Selected work excerpt; client-sensitive content redacted.

Water distribution math review

My role Math SME review

I checked a technical training proof against its answer key and recorded six specific corrections to units, wording, and calculations.

ORIGINAL PROOF AND MATH REVIEW NOTES



SELECTED REVIEW COMMENTS

01 Arithmetic checked against the key
02 Units clarified
03 Table labels reconciled with the question
04 Requested quantity made explicit
05 Calculation corrected
06 Dosage reasoning verified

Math review notes. Selected work excerpt; client-sensitive content redacted.

Five-grade standards alignment prompt

My role Prompt author

I wrote a five-grade check for deciding whether a math question truly fits a target standard. It tests number limits and representations before looking at nearby grades.

ORIGINAL WORK FILE EXCERPT

I will provide:

A math question.

A target grade level.

The full set of Common Core Math standards for: • the target grade, • one grade below and one grade above, • two grades below and two grades above.

Your task:

Step 1: Check the target grade first. • Determine if the question fully aligns with one standard in that grade. • Match must obey every constraint in the standard's text (number limits, operation types, number types, representation requirements, exact verbs, and problem intent). • If it fully matches, stop and report that standard.

Step 2: If no full alignment in the target grade, check one grade below and one grade above. • Apply the same strict compliance check. • If a match is found, stop and report that standard.

Step 3: If still no match, check two grades below and two grades above. • Apply the same strict compliance check. • If a match is found, stop and report that standard.

Step 4: If no standard across this five-grade band fully matches, respond that the question does not align.

Strict alignment rules:

DOK 4 tutor-task prompt

My role Prompt author

I set rules for an extended-thinking task a student can complete with an AI tutor. The prompt asks for decisions, testing, revision, and reflection while staying inside the grade-level standard.

ORIGINAL WORK FILE EXCERPT

Craft a single DOK 4 student-facing question

Explain DOK levels in plain terms (for your planning, not to show students). Make it distinct and go beyond DOK 1-3:

DOK 1: recall a fact or do a routine step.

DOK 2: apply skills or concepts in multiple steps.

DOK 3: reason strategically, justify thinking, and solve a non-routine problem.

DOK 4: plan and carry out a task that requires the student to use extended thinking, such as gather data or research, test ideas, make decisions, revise, and reflect, though only anything they can do with a tutor (don't mention this). If you ask the student to test something, be very specific on what they are testing and how to test it. For example, don't ask " looking at samples your tutor will provide.", instead, provide the sample along with the question.

It should assess only and exactly the following standard, as much of it as possible with the output constraints. Pay attention to the first number, and make it appropriate for that grade in terms of math level and reading level. All questions should have context and mimic something that can happen in the real world as closely as possible: [Standard code + standard language]

Now write exactly one DOK 4 question that includes something similar to one of the following activities, though this is not an exhaustive list:

creating and labeling hand-drawn models, planning and running tutor-given number trials and recording results, testing two paper

Math question gap-filling prompt

My role Prompt author

I wrote guardrails for generating standards-aligned math questions when a bank has a gap. The questions have to use realistic quantities, clear units, and enough information to stand on their own.

ORIGINAL WORK FILE EXCERPT

Action Input: <final answer with units if applicable>

* Focus on this learning standard: **{{Full standard sentence}}**. The math must make sense in the scenario. Use realistic magnitudes, correct units, and consistent constraints. Label all quantities and units. Avoid impossible values and confirm totals and bounds are consistent before outputting any question.

- Clear, well-structured, and self-contained (all necessary information is included).

- Free from the requirement of images, graphs, or visual aids.

- Appropriate for a student with {{learning_standard_grade }} grade skills. It's very important to only use appropriate language for that grade. Make sure you use as simple language as possible, since this is a math assessment, not one of understanding context.

Action Input: <final answer with units if applicable>

* Focus on this learning standard: Work with the basic measure of perimeter and area. The math must make sense in the scenario. Use realistic magnitudes, correct units, and consistent constraints. Label all quantities and units. Avoid impossible values and confirm totals and bounds are consistent before outputting any question.

- Clear, well-structured, and self-contained (all necessary information is included).

- Free from the requirement of images, graphs, or visual aids.

Multiple-integrals test-bank blueprint

My role Blueprint and math review

I planned six multiple-integrals lessons by exercise family, DOK level, objective, and graphic need. The blueprint organized 48 exercise sets and 528 questions for writing.

ORIGINAL WORK FILE EXCERPT

Chapter 14 — Multiple Integrals

6 lessons | 48 exercise sets | 528 individual questions

Set | Qs | DOK | Obj | Summary

14.1A | 8 | 1 | 14.1.1 | Identify components of a double Riemann sum.

14.1B | 10 | 2 | 14.1.1 | Write or approximate double Riemann sums over rectangular regions.

14.1C | 14 | 2 | 14.1.2 | Evaluate iterated double integrals.

14.1D | 12 | 2 | 14.1.3 | Reverse the order of integration for a double integral over a bounded region.

14.1E | 12 | 2 | 14.1.3 | Set up, but do not evaluate, a double integral over a bounded region.

Calculus item math and dependency audit

My role Math and item-quality review

I audited a 25-question calculus bank for math, answer logic, LaTeX, and variant dependencies. The review calls out six blocking errors and gives writers exact fixes.

ORIGINAL WORK FILE EXCERPT

5.3 Question Bank — Full Math & Dependency Audit (25 Questions)

Status legend: OK = no changes needed; Minor = cosmetic fix; Fix… = content issue; FIX (caps) = blocker (math or rendering).

Summary of blockers

• Q3 — interval typo: [0, 2/π] must be [0, 2π].

• Q7 — LaTeX for √(x+1) is broken: rendered as √x + 1 in choices A, B, D and solution.

• Q9 — Answer line has a broken bold/fraction wrapper ("**\frac{**4}{3}").

• Q11 — closing brace typo \right} instead of \right).

• Q20 — stem limit expression is malformed (exponent '2 · · 1/n', Σ swallowed by lim subscript); leftover author note 'ONLY 2 VARIANTS FOR THIS'.

• Q25 — extensive LaTeX never rendered as math (\\sqrt, \thinsp, \right} shown literally).

Recurring highlight issues

• Answer lines should not be highlighted. Violations: Q5, Q14, Q15, Q19, Q20, Q21, Q24 (and the dollar amount in Q24 only incidentally contains the variable digits).

• Same value highlighted in different colors for the same dependency: Q6 (2 as plateau height — yellow vs green), Q12 (the '2' of 2−4x

Precalculus quiz review log

My role Editorial and math review

I tracked changes to 17 precalculus quiz questions through QC and SME review. The log shows which feedback on number lines, notation, and solution steps was addressed.

WORKBOOK EXCERPT / Sheet1

Question Number	QC Comment	SME Comment	QC Comment 2	SME Response 2
1	Followed pedagogy.			
2	Not followed. Additional details, such as domain constraints are provided.	I deleted the step from the explanation that listed which numbers x isn't allowed to be,	In step 6:Please remove the statement "We must check our solutions against the	Removed extra statement from the explanation.
3	Not followed. The number line is missing, and terminology such as critical points,	Added a number line picture to the explanation and replaced "critical values," and	In the number line, show the points selected for testing. Also, add a number line at the	Added the number line showing the intervals and test points, and updated the explanation to
4	Followed pedagogy.			

Piecewise-function item repair

My role Item editing and math review

I revised a 27-item piecewise-function set to make graph choices and endpoint inclusion precise. The excerpt shows how the answer options differ at the boundaries.

ORIGINAL WORK FILE EXCERPT

Item 1 of 27

Question update (just the final question):

Question update:

Select the graph that matches the piecewise function.

[Keep the function as shown currently.]

Multiple Choice (display each option as a graph on -10 ≤ x ≤ 10 and -10 ≤ y ≤ 10)

A. f(x) = { x^3 if x < -1 ; -2 if -1 ≤ x < 4 ; sqrt(x) if x ≥ 4 }

B. f(x) = { x^3 if x ≤ -1 ; -2 if -1 < x < 4 ; sqrt(x) if x ≥ 4 }

C. f(x) = { x^3 if x < -1 ; -2 if -1 ≤ x ≤ 4 ; sqrt(x) if x > 4 }

D. f(x) = { x^3 if x < -1 ; 2 if -1 ≤ x < 4 ; sqrt(x) if x ≥ 4 }

Answer: A

Algebra II and geometry assessment blueprint

My role Assessment planning and SME review

I helped plan 30-item geometry and algebra forms, balancing item types and mathematical domains so the assessment covers more than routine multiple choice.

WORKBOOK EXCERPT / Item Types

	BOY|MY|EOY Assessment	Multiple Choice	Technology Enhanced	Math Cloze (Fill in the blank)
Grade	Total Number of Items	Number of Items	Number of Items	Number of Items
Course 1	30.0	15.0		5.0
Course 2	30.0	15.0		5.0
Course 3	30.0	15.0		10.0
Algebra 1	30.0	12.0	9.0	4.0
Geometry	30.0	15.0	10.0	5.0
Algebra 2	30.0	15.0	10.0	5.0

Assessment blueprint. Selected work excerpt; client-sensitive content redacted.

Residual-plot student activity

My role Editorial direction and math SME review

I helped shape a statistics activity that moves from a scatterplot to a regression line and residual plot. Students use the evidence to decide whether a new point is influential.

ORIGINAL WORK FILE EXCERPT

SOURCE STRUCTURE / CONTENT WITHHELD

01

Scatterplot

02

Influential point

03

Regression line

04

Residual plot

Activity excerpt. Selected work excerpt; client-sensitive content redacted.

Taylor-series item review

My role Math SME and item review

I reviewed higher-ed calculus items on binomial and Taylor series, including learning objectives and answer choices. The excerpt shows the section structure and the questions under review.

ORIGINAL WORK FILE EXCERPT

Chapter 10: Power Series

Section 10.4 Working with Taylor Series

Binomial Series

Learning Objectives

Use a binomial series

Functions Expressed as Taylor Series

Learning Objectives

Find the Taylor expansion of common functions

Find the Taylor expansion of common functions

None of these.

Find the radius of convergence in a Taylor Series

The function f(x) = is a Taylor series centered at . Which of the following is the correct expression for the Taylor series and radius of convergence?

Florida B.E.S.T. to CCSS crosswalk

My role Crosswalk author

I compared grades 3–5 B.E.S.T. benchmarks with CCSS standards and recorded the strength and limits of each match. The notes make the alignment decision visible.

ORIGINAL WORK FILE EXCERPT

Grade 3 measurement and data crosswalk

3.MD.1 Tell and write time to the nearest minute; solve time word problems → MA.3.M.2.1 Solve problems involving elapsed time

Alignment: Full. B.E.S.T. matches closely — elapsed time calculation emphasized.

3.MD.2 Measure and estimate liquid volumes and masses of objects → MA.3.M.1.1 Estimate and measure liquid volumes/mass

Alignment: Full. Practical applications in measurement problems.

3.MD.3 Draw scaled picture and bar graphs to represent data → MA.3.DP.1.1 Create scaled bar graphs and pictographs

Alignment: see note. Strong alignment. B.E.S.T. focuses heavily on scaled representation.

3.MD.4 Generate data and represent with scaled bar graphs, pictographs, line plots → MA.3.DP.1.2 Display and interpret data sets using line plots

Alignment: Full. Line plots with fractional values begin here.

Financial Algebra quality-rubric review

My role Editorial and SME evidence review

I checked a Financial Algebra program against a Texas instructional-materials rubric. The evidence log ties specific course materials to each review criterion.

ORIGINAL WORK FILE EXCERPT

Texas instructional materials quality rubric — selected evidence

Standard | Descriptor | Citations

1.1a | Materials include a scope and sequence outlining the TEKS, ELPS, and concepts taught in the course. ; | This standard is not covered within the components of Financial Algebra: Advanced Algebra with Financial Applications ©(2026)

1.1b | Materials include suggested pacing (pacing guide/calendar) to support effective implementation for various instructional calendars (e.g., varying numbers of instructional days – 165, 180, 210). ; | Teacher lesson plans: Lesson 1 - 11; Section "Pacing Guides"

1.1c | Materials include an explanation for the rationale of unit order as well as how concepts to be learned connect throughout the course. ; | TE: Chapter Overviews, pages 2, 70, 168, 244, 356, 406, 482, 568, 654, 736, 802

Grade 4 standards remapping audit

My role Standards and item review

I reviewed 386 flagged grade 4 rows where the assigned standard changed. The workbook preserves the tested skill, old code, new code, and rationale.

WORKBOOK EXCERPT / Flagged Grade 4 Rows

TI #	Tested Skill	Original Standard Code	Updated Standard Code
111	multiply and add to find total cost from table	4.MA.NSW.5	4.MA.NSF.3
118	compute difference in minutes between fractional and whole hours	4.MA.M.4	4.MA.M.1
119	add fractions with unlike denominators and simplify	4.MA.NSF.2	5.MA.NSN.4
123	identify place value of digit in decimal number	4.MA.NSF.1	6.MA.NNR.1
124	perform long division of whole numbers to find exact quotient	4.MA.NSW.3	5.MA.NSN.4

Standards audit excerpt. Selected work excerpt; client-sensitive content redacted.

Math-content census across technical titles

My role Math SME scoping

I inventoried pages with math across technical training books so the team could size its review effort. This is the page-level count behind the staffing estimate.

WORKBOOK EXCERPT / Technical titles

Treatment Grade 1		Distribution Grade 1	
Chapter	Number of Pages with Math	Chapter	Number of Pages with Math
1	5	2	47
2	46	3	3
5	1	6	5
Total Pages Estimate	425	15	1
Percentage of Book	0.1223529411764706	Total Pages Estimate	400
		Percentage of Book	0.14

Math inventory excerpt. Selected work excerpt; client-sensitive content redacted.

Grade 7 circle-graph lesson sample

My role Editorial direction and math SME review

I provided editorial and math review on a grade 7 lesson that moves from reading a circle graph to using proportions and sector angles. The excerpt shows the student-facing progression.

ORIGINAL WORK FILE EXCERPT

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

Lesson Title: The County Fair

Let’s use proportional reasoning to make sense of circle graphs.

Warm-Up

1. Here is a circle graph showing a family’s budget at a county fair. What do you notice? What do you wonder?

The percents add to 100.

The largest piece of the graph represents the greatest percentage, and the smallest piece of the graph represents the smallest percentage.

How can you determine the size of each section of the circle graph?

How do the percentages relate to amounts of money in a budget?

Activity 1: Setting Up

Circle graphs visually display data as parts of a whole. In a circle graph, the parts add up to 100 percent.

The diagram shows the percent of space occupied by different activities at the fair.

Introducing multiplication unit design

My role Editorial direction and math SME review

I helped guide a multiplication unit from equal groups and arrays through skip counting and word problems. The sequence shows how early concepts build before independent practice.

ORIGINAL WORK FILE EXCERPT

Unit: Introducing Multiplication
Recall Prior Knowledge
Task Model
Equal Groups
Direct Instruction
Guided Practice - 3 Part Item
Guided Practice - 4 Part Item
Independent Practice (Option 1)
Independent Practice (Option 2)
Understanding Arrays
Multiplying by 2
Multiplying by 5
Multiplying by 10
Synthesis & Quiz

Ratios and percents tutor evidence map

My role AI tutor math evaluation

I mapped productive and problematic reasoning about ratios and percents using 31 anchor questions and 95 eligible tutor conversations. The evidence sets boundaries for what the tutor can infer.

ORIGINAL WORK FILE EXCERPT

AUTONOMOUS CONVERSATION-INFORMED FACET MAP

Map 26: Ratios, Percents, and Proportional Reasoning

Grades 7–8 | Autonomous, conversation-informed facet map

Standards: 7.MA.NNR.2 • 8.MA.NNR.2

Working-final autonomous set: 5 Map-26-owned productive facets and 7 problematic facets. Equivalent ratio/rate scaling, percent-of-quantity, inverse part–whole reasoning, and multistep operation sequencing reuse already validated facets from Maps 24, 25, and 30. This keeps the map focused on small beliefs: what the two ratio terms refer to, how a part–part ratio relates to a whole, how a combined total is partitioned by ratio units, which quantity is the reference whole for a percent, and why percent change is relative to the original amount.

Scope and evidence base

7.MA.NNR.2: Recognize the form of a number appropriate for use in a given situation and apply ratio, percent, and proportion in a variety of situations.

8.MA.NNR.2: Recognize the form of a number appropriate for use in a given situation and apply ratio, percent, and proportion in a variety of situations.

Use of generated questions: Generated questions were used to discover recurring student reasoning and refine evidence boundaries.

Expected-value student activity

My role Editorial direction and math SME review

I reviewed an expected-value activity built around an extended-warranty decision. Students use probabilities and payouts to explain what the expected value means for a consumer.

ORIGINAL WORK FILE EXCERPT

SOURCE STRUCTURE / CONTENT WITHHELD

01

Warranty scenario

02

Possible outcomes

03

Expected value

04

Consumer decision

Math alt-text rules

My role Editorial direction and accessibility SME review

I helped guide the rules for describing equations, tables, and graphs in math content. The examples show how spoken math can stay precise without overloading the reader.

ORIGINAL WORK FILE EXCERPT

General Alt Text Rules:

Always spell out math symbols such as negatives, operations, equality, etc.

: y equals negative 4 fifths x plus 3

$10.50: 10 dollars and 50 cents

As above, spell out fractions. Also spell out decimals using mathematical language.

4.56: 4 and 56 hundredths

Do not use quotation marks or parentheses unless the client asks for these.

A student says, “The sum is 20.” → A student says, the sum is 20.

Cost ($): Cost in dollars

For ordered pairs, do not use parentheses. Spell out the comma.

(4, –2): the point 4 comma negative 2

Separate numbers and letters used in sequence.

3x: 3 x

: triangle A B C

Lesson production and review schedule

My role Production planning and editorial direction

I scheduled 96 lessons across seven freelancers, from parent-item drafting through review, variants, and final approval. The plan makes each handoff and its volume visible.

WORKBOOK EXCERPT / Freelancer Chapter Map

Freelancer	Chapter block	Actual lesson count	Parent questions	Variants
A	Chapter 1, Chapter 2	11	275	1650
B	Chapter 3, Chapter 4	16	400	2400
C	Chapter 5, Chapter 6	11	275	1650
D	Chapter 7, Chapter 8	11	275	1650
E	Chapter 9, Chapter 10	15	375	2250
F	Chapter 11, Chapter 12, Chapter 13	18	450	2700
G	Chapter 14, Chapter 15	14	350	2100

Production schedule excerpt. Selected work excerpt; client-sensitive content redacted.